Syllabus Details

Syllabus ID:	13152
Syllabus Name:	**Fundamental Game Development_Phát triển game cơ bản**
Course Name English:	**Fundamental Game Development**
Subject Code:	**FGU301**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	45h (60 sessions) contact hours + 1h final exam + 104h self-study
Pre-Requisite:	PRO192
Description:	This comprehensive Basic Unity Training Program is designed to equip participants with essential skills in Unity game development, empowering them to create engaging and interactive 2D and 3D games. Building upon strong foundation in C# programming, the course delves into advanced topics such as 3D game development, physics, animations, and user interface design. Through a combination of theoretical lectures, hands-on labs, and collaborative projects, participants will gain practical experience and confidence in using Unity to bring their game ideas to life. Key Learning Outcomes: - Master C# Programming: Gain a deep understanding of C# syntax, data structures, algorithms, and object-oriented programming principles. - Navigate the Unity Environment: Become proficient in using Unity's interface and tools for game development. - Develop 3D Games: Learn the fundamentals of 3D game development, including rendering, lighting, animations, and physics. - Create Engaging Game Worlds: Master the art of designing game objects, animations, and physics components to create immersive and interactive game environments. - Build User-Friendly Interfaces: Design intuitive and visually appealing user interfaces that enhance the player experience. - Develop Games from Start to Finish: Complete hands-on projects, applying your knowledge to create your own small games and interactive applications. - Collaborate Effectively: Work collaboratively with teammates to develop game projects, fostering effective communication and teamwork skills.
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to http://flm.fpt.edu.vn for up-to-date course information
Tools:	- Internet - Visual Studio 2019 or later (C# programming language) - Unity Hub and Unity 5 (version 2021.3 LTS or later) - Minimum hardware requirement: Windows 10, 64-bit versions, CPU: X64, RAM: 8GB - Students use Unity with their personal email for educational purposes only
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

5 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Unity Official Document	Unity Technologies	Unity Technologies				True	False	True	https://docs.unity3d.com/Manual/index.html

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
2	Microsoft Learn: C#	Microsoft	Microsoft				True	False	True	https://learn.microsoft.com/en-us/dotnet/csharp/ of-csharp/ overview
3	C# Programming for Unity Game Development Specialization	University of Colorado System	Coursera				False	False	True	https://www.coursera.org/specializations/prograr unity-game-development
4	Unity Learn Platform	Unity Technologies	Unity Technologies				False	False	True	https://learn.unity.com
5	Unity Official Channel	Unity Technologies	Youtube				False	False	True	https://www.youtube.com/@unity

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand C# Programming Fundamentals: Syntax and Data Types, Control Flow, Functions and Methods, Classes and Objects, Debugging
2	CLO2	Understand C# data structures (lists, stacks, queues, trees) and algorithms (sorting, searching); Event Handling
3	CLO3	Understand the Unity interface and basic tools for game development
4	CLO4	Understand 3D Game Development; Rendering and Lighting; Animations and Physics
5	CLO5	Create and manipulate game objects, animations, and physics components
6	CLO6	Design user interfaces and handle user input effectively
7	CLO7	Develop small games and interactive applications using Unity
8	CLO8	Collaborate effectively in a team environment for game development projects.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Chapter 1: Introduction to Unity 1.1 Unity interface 1.2 Scene, game object, component 1.3 Create a simple 2D or 3D project	Offline	CLO3, CLO5	IT	- Chapter 1 slide and online materials	FGU301_Slide	Read the slide of Chapter 1: Introduction to Unity	
2	Chapter 1: Introduction to Unity (cont.) 1.1 Unity interface 1.2 Scene, game object, component 1.3 Create a simple 2D or 3D project	Offline	CLO3, CLO5	IT	- Chapter 1 slide and online materials	link2-1	Read the slide of Chapter 1: Introduction to Unity	
3	Chapter 1: Introduction to Unity (cont.) 1.1 Unity interface 1.2 Scene, game object, component 1.3 Create a simple 2D or 3D project	Offline	CLO3, CLO5	IT	- Chapter 1 slide and online materials	link3-1	Read the slide of Chapter 1: Introduction to Unity	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
4	Chapter 2: C# Fundamentals 2.1 C# script component 2.2 Variables, data types, operators 2.3 Control flow (if-else, loops)	Offline	CLO 1	IT	- Chapter 2 slide and online materials	link4-1	Read the slide of Chapter 2: C# Fundamentals	
5	Chapter 2: C# Fundamentals (cont.) 2.1 C# script component 2.2 Variables, data types, operators 2.3 Control flow (if-else, loops)	Offline	CLO 1	IT	- Chapter 2 slide and online materials	link5-1	Read the slide of Chapter 2: C# Fundamentals	
6	Chapter 2: C# Fundamentals (cont.) 2.1 C# script component 2.2 Variables, data types, operators 2.3 Control flow (if-else, loops)	Offline	CLO 1	IT	- Chapter 2 slide and online materials	link6-1	Read the slide of Chapter 2: C# Fundamentals	
7	Chapter 2: C# Fundamentals (cont.) 2.4 Classes, objects, methods, properties 2.5 Inheritancea, polymorphism, encapsulation 2.6 Arrays, lists, dictionaries	Offline	CLO2	IT	- Chapter 2 slide and online materials	link7-1	Read the slide of Chapter 2: C# Fundamentals	
8	Chapter 2: C# Fundamentals (cont.) 2.4 Classes, objects, methods, properties 2.5 Inheritancea, polymorphism, encapsulation 2.6 Arrays, lists, dictionaries	Offline	CLO2	IT	- Chapter 2 slide and online materials	link8-1	Read the slide of Chapter 2: C# Fundamentals	
9	Chapter 2: C# Fundamentals (cont.) 2.4 Classes, objects, methods, properties 2.5 Inheritancea, polymorphism, encapsulation 2.6 Arrays, lists, dictionaries	Offline	CLO2	IT	- Chapter 2 slide and online materials	link9-1	Read the slide of Chapter 2: C# Fundamentals	
10	Hands-on 1: Project 1 - Simple 2D project Create a basic platforming game, focusing on character movement, collision detection, and scoring. Quiz 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO7, CLO8	U			Review Chapter 1, 2 Implement project 1 Do Quiz 1	
11	Hands-on 1 (cont.): Project 1 - Simple 2D project Create a basic platforming game, focusing on character movement, collision detection, and scoring. Quiz 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO7, CLO8	U			Review Chapter 1, 2 Implement project 1 Do Quiz 1	
12	Hands-on 1 (cont.): Project 1 - Simple 2D project Create a basic platforming game, focusing on character movement, collision detection, and scoring. Quiz 1	Offline	CLO1, CLO2, CLO3, CLO5, CLO7, CLO8	U			Review Chapter 1, 2 Implement project 1 Do Quiz 1	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
13	Chapter 3: Introduction to 2D and 3D Graphics in Unity 3.1 2D/3D Graphics: Sprites, meshes, materials 3.2 Mecanim and Animation Controller.	Offline	CLO3, CLO4	IT	- Chapter 3 slide and online materials	link13-1	Read the slide of Chapter 3: Introduction to 2D and 3D	
14	Chapter 3: Introduction to 2D and 3D Graphics in Unity (cont.) 3.1 2D/3D Graphics: Sprites, meshes, materials 3.2 Mecanim and Animation Controller.	Offline	CLO3, CLO4	IT	- Chapter 3 slide and online materials	link14-1	Read the slide of Chapter 3: Introduction to 2D and 3D	
15	Chapter 3: Introduction to 2D and 3D Graphics in Unity (cont.) 3.1 2D/3D Graphics: Sprites, meshes, materials 3.2 Mecanim and Animation Controller.	Offline	CLO3, CLO4	IT	- Chapter 3 slide and online materials	link15-1	Read the slide of Chapter 3: Introduction to 2D and 3D	
16	Chapter 3: Introduction to 2D and 3D Graphics in Unity (cont.) 3.3 Unity GUI 3.4 Introduce UI event handling and user input	Offline	CLO6	IT	- Chapter 3 slide and online materials	link16-1	Read the slide of Introduction to 2D and 3D Graphics in Unity	
17	Chapter 3: Introduction to 2D and 3D Graphics in Unity (cont.) 3.3 Unity GUI 3.4 Introduce UI event handling and user input	Offline	CLO6	IT	- Chapter 3 slide and online materials	link17-1	Read the slide of Introduction to 2D and 3D Graphics in Unity	
18	Chapter 3: Introduction to 2D and 3D Graphics in Unity (cont.) 3.3 Unity GUI 3.4 Introduce UI event handling and user input	Offline	CLO6	IT	- Chapter 3 slide and online materials	link18-1	Read the slide of Introduction to 2D and 3D Graphics in Unity	
19	Chapter 4: Unity's Physics Engine 4.1 Rigidbodies, colliders 4.2 Forces, torque	Offline	CLO4	IT	- Chapter 4 slide and online materials	link19-1	Read the slide of Chapter 4: Unity's Physics Engine	
20	Chapter 4: Unity's Physics Engine (cont.) 4.1 Rigidbodies, colliders 4.2 Forces, torque	Offline	CLO4	IT	- Chapter 4 slide and online materials	link20-1	Read the slide of Chapter 4: Unity's Physics Engine	
21	Chapter 4: Unity's Physics Engine (cont.) 4.1 Rigidbodies, colliders 4.2 Forces, torque	Offline	CLO4	IT	- Chapter 4 slide and online materials	link21-1	Read the slide of Chapter 4: Unity's Physics Engine	
22	Hands-on 2: Project 1 - 2D Platformer Expansion Expand the platformer game with levels, enemies, and power-ups. Quiz 2	Offline	CLO3, CLO4, CLO6, CLO7, CLO8	U			Reveiw Chapter 3, 4 Implement project 1 (cont.) Do Quiz 2	
23	Hands-on 2 (cont.): Project 1 - 2D Platformer Expansion Expand the platformer game with levels, enemies, and power-ups. Quiz 2	Offline	CLO3, CLO4, CLO6, CLO7, CLO8	U			Reveiw Chapter 3, 4 Implement project 1 (cont.) Do Quiz 2	
24	Hands-on 2 (cont.): Project 1 - 2D Platformer Expansion (cont.) Expand the platformer game with levels, enemies, and power-ups. Quiz 2	Offline	CLO3, CLO4, CLO6, CLO7, CLO8	U			Reveiw Chapter 3, 4 Implement project 1 (cont.) Do Quiz 2	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
25	Assignment 1 assessment: Project 1 - 2D Platformer Expansion Project Presentations and Feedback	Offline	CLO4, CLO6, CLO7, CLO8	TU			Present projects 1	
26	Assignment 1 assessment (cont.): Project 1 - 2D Platformer Expansion Project Presentations and Feedback	Offline	CLO4, CLO6, CLO7, CLO8	TU			Present projects 1	
27	Assignment 1 assessment (cont.): Project 1 - 2D Platformer Expansion Project Presentations and Feedback	Offline	CLO4, CLO6, CLO7, CLO8	TU			Present projects 1	
28	Chapter 5: Input Handling 5.1 Keyboard, mouse input 5.2 Unity Input System 5.3 Unity Input Manager	Offline	CLO3	TU	- Chapter 5 slide and online materials	link28-1	Read the slide of Chapter 5: Input Handling	
29	Chapter 5: Input Handling (cont.) 5.1 Keyboard, mouse input 5.2 Unity Input System 5.3 Unity Input Manager	Offline	CLO3	TU	- Chapter 5 slide and online materials	link29-1	Read the slide of Chapter 5: Input Handling	
30	Chapter 5: Input Handling (cont.) 5.1 Keyboard, mouse input 5.2 Unity Input System 5.3 Unity Input Manager	Offline	CLO3	TU	- Chapter 5 slide and online materials	link30-1	Read the slide of Chapter 5: Input Handling	
31	Chapter 6: Interactive Programming 6.1 Event Handling: Event-driven programming 6.2 Coroutines: Time-based execution of code 6.3 Serialization: Saving and loading game data	Offline	CLO3	IT	- Chapter 6 slide and online materials	link31-1	Read the slide of Chapter 6: Interactive Programming	
32	Chapter 6: Interactive Programming (cont.) 6.1 Event Handling: Event-driven programming 6.2 Coroutines: Time-based execution of code 6.3 Serialization: Saving and loading game data	Offline	CLO3	IT	- Chapter 6 slide and online materials	link32-1	Read the slide of Chapter 6: Interactive Programming	
33	Chapter 6: Interactive Programming (cont.) 6.1 Event Handling: Event-driven programming 6.2 Coroutines: Time-based execution of code 6.3 Serialization: Saving and loading game data	Offline	CLO3	IT	- Chapter 6 slide and online materials	link33-1	Read the slide of Chapter 6: Interactive Programming	
34	Chapter 7: Advanced C# 7.1 Delegates, events, coroutines 7.2 Unity's component-based architecture (ECS)	Offline	CLO1, CLO2	IT	- Chapter 7 slide and online materials	link34-1	Read the slide of Chapter 7: Advanced C#	
35	Chapter 7: Advanced C# (cont.) 7.1 Delegates, events, coroutines 7.2 Unity's component-based architecture (ECS)	Offline	CLO1, CLO2	IT	- Chapter 7 slide and online materials	link35-1	Read the slide of Chapter 7: Advanced C#	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
36	Chapter 7: Advanced C# (cont.) 7.1 Delegates, events, coroutines 7.2 Unity's component-based architecture (ECS)	Offline	CLO1, CLO2	IT	- Chapter 7 slide and online materials	link36-1	Read the slide of Chapter 7: Advanced C#	
37	Hands-on 3: Project 2 - Create Simple 3D Puzzle Game: Design a puzzle game using data structures and algorithms. Quiz 3	Offline	CLO1, CLO3, CLO7, CLO8	U			Review Chapter 5, 6, 7 Implement project 2 Do Quiz 3	
38	Hands-on 3 (cont.): Project 2 - Create Simple 3D Puzzle Game: Design a puzzle game using data structures and algorithms. Quiz 3	Offline	CLO1, CLO3, CLO7, CLO8	U			Review Chapter 5, 6, 7 Implement project 2 Do Quiz 3	
39	Hands-on 3 (cont.): Project 2 - Create Simple 3D Puzzle Game: Design a puzzle game using data structures and algorithms. Quiz 3	Offline	CLO1, CLO3, CLO7, CLO8	U			Review Chapter 5, 6, 7 Implement project 2 Do Quiz 3	
40	Assignment 2 assessment: Project 2 - 3D game Project Presentations and Feedback	Offline	CLO1, CLO2, CLO3, CLO7, CLO8	TU			Present projects 2	
41	Assignment 2 assessment (cont.): Project 2 - 3D game Project Presentations and Feedback	Offline	CLO1, CLO2, CLO3, CLO7, CLO8	TU			Present projects 2	
42	Assignment 2 assessment (cont.): Project 2 - 3D game Project Presentations and Feedback	Offline	CLO1, CLO2, CLO3, CLO7, CLO8	TU			Present projects 2	
43	Chapter 8: Unity Audio 8.1 Importing sound effects 8.2 Playing sound effects	Offline	CLO3	TU	- Chapter 8 slide and online materials	link43-1	Read the slide of Chapter 8: Unity Audio	
44	Chapter 8: Unity Audio (cont.) 8.1 Importing sound effects 8.2 Playing sound effects	Offline	CLO3	TU	- Chapter 8 slide and online materials	link44-1	Read the slide of Chapter 8: Unity Audio	
45	Chapter 8: Unity Audio (cont.) 8.1 Importing sound effects 8.2 Playing sound effects	Offline	CLO3	TU	- Chapter 8 slide and online materials	link45-1	Read the slide of Chapter 8: Unity Audio	
46	Chapter 9: Game Logic and Scoring Systems 9.1 Game Logic and Scoring Systems 9.2 Using AI in Unity games 9.3 Basic networking concepts in Unity	Offline	CLO4, CLO5	IT	- Chapter 8 slide and online materials	link46-1	Read the slide of Chapter 9: Game Logic and Scoring Systems	
47	Chapter 9: Game Logic and Scoring Systems (cont.) 9.1 Game Logic and Scoring Systems 9.2 Using AI in Unity games 9.3 Basic networking concepts in Unity	Offline	CLO4, CLO5	IT	- Chapter 8 slide and online materials	link47-1	Read the slide of Chapter 9: Game Logic and Scoring Systems	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
48	Chapter 9: Game Logic and Scoring Systems (cont.) 9.1 Game Logic and Scoring Systems 9.2 Using AI in Unity games 9.3 Basic networking concepts in Unity	Offline	CLO4, CLO5	IT	- Chapter 8 slide and online materials	link48-1	Read the slide of Chapter 9: Game Logic and Scoring Systems	
49	Chapter 9: Game Logic and Scoring Systems (cont.) 9.1 Game Logic and Scoring Systems 9.2 Basic AI concept in Unity game 9.3 Basic networking concepts in Unity	Offline	CLO4, CLO5	IT	- Chapter 9 slide and online materials	link49-1	Read the slide of Chapter 9: Game Logic and Scoring Systems	
50	Chapter 9: Game Logic and Scoring Systems (cont.) 9.1 Game Logic and Scoring Systems 9.2 Basic AI concept in Unity game 9.3 Basic networking concepts in Unity	Offline	CLO4, CLO5	IT	- Chapter 9 slide and online materials	link50-1	Read the slide of Chapter 9: Game Logic and Scoring Systems	
51	Chapter 9: Game Logic and Scoring Systems (cont.) 9.1 Game Logic and Scoring Systems 9.2 Basic AI concept in Unity game 9.3 Basic networking concepts in Unity	Offline	CLO4, CLO5	IT	- Chapter 9 slide and online materials	link51-1	Read the slide of Chapter 9: Game Logic and Scoring Systems	
52	Hands-on 4: Project 2 - Implement 3D game	Offline	CLO7, CLO8	U			Review Chapter 8, 9 Implement project 2 (cont.)	
53	Hands-on 4 (cont.): Project 2 - Implement 3D game	Offline	CLO7, CLO8	U			Review Chapter 8, 9 Implement project 2 (cont.)	
54	Hands-on 4 (cont.): Project 2 - Implement 3D game	Offline	CLO7, CLO8	U			Review Chapter 8, 9 Implement project 2 (cont.)	
55	Final Project Presentations and Feedback	Offline	CLO7, CLO8	U			Present final project	
56	Final Project Presentations and Feedback (cont.)	Offline	CLO7, CLO8	U			Present final project	
57	Final Project Presentations and Feedback (cont.)	Offline	CLO7, CLO8	U			Present final project	
58	Final Project Presentations and Feedback (cont.)	Offline	CLO7, CLO8	U			Present final project	
59	Final Project Presentations and Feedback (cont.)	Offline	CLO7, CLO8	U			Present final project	
60	Final Project Presentations and Feedback (cont.)	Offline	CLO7, CLO8	U			Present final project	

0 Constructive question(s)

3 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment (Lab)	on-going	2	40.0%	>0	90'	Assignment 1: CLO4, CLO6, CLO7, CLO8 Assignment 2: CLO1, CLO2, CLO3, CLO7, CLO8	Theory present or project		- Assignment 1 cover content: Chapter 1, 2, 3, 4 - Assignment 2 cover content: Chapter 3, 4, 5, 6, 7	- Guided by instructor - Application or Theory - Work in group - Presentation	Instructor has responsibility to review the test for students after graded.
2	Progress test (Quiz)	on-going	3	30.0%	>0	10'-30'	Quiz 1: CLO1, CLO2, CLO3, CLO5, Quiz 2: CLO3, CLO4, CLO6 Quiz 3: CLO1, CLO3	Multiple choices Marked by Computer or a suitable format	30	Quiz 1 cover content: Chapter 1, 2 Quiz 2 cover content: Chapter 3, 4 Quiz 3 cover content: Chapter 5, 6, 7	by exam board, using computer	Instruction and schedules for Progress tests must be presented in the Course Implementation Plan approved by director of the campus. Progress test must be taken right after the last lectures of required material. Instructor has responsibility to review the test for students after graded.
3	Final exam	Final exam	1	30.0%	4	60'	CLO1 - CLO6	Multiple choices Marked by Computer	50	concepts, algorithms; all studied chapters	by exam board, using computer	The exam questions must be updated or different at least 70% to the previous ones.